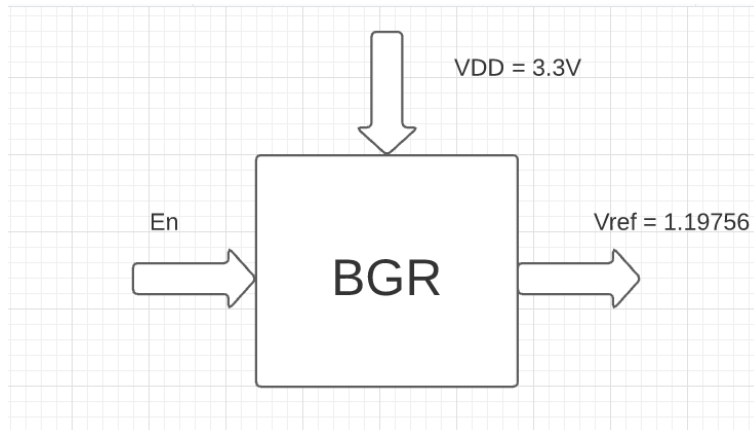
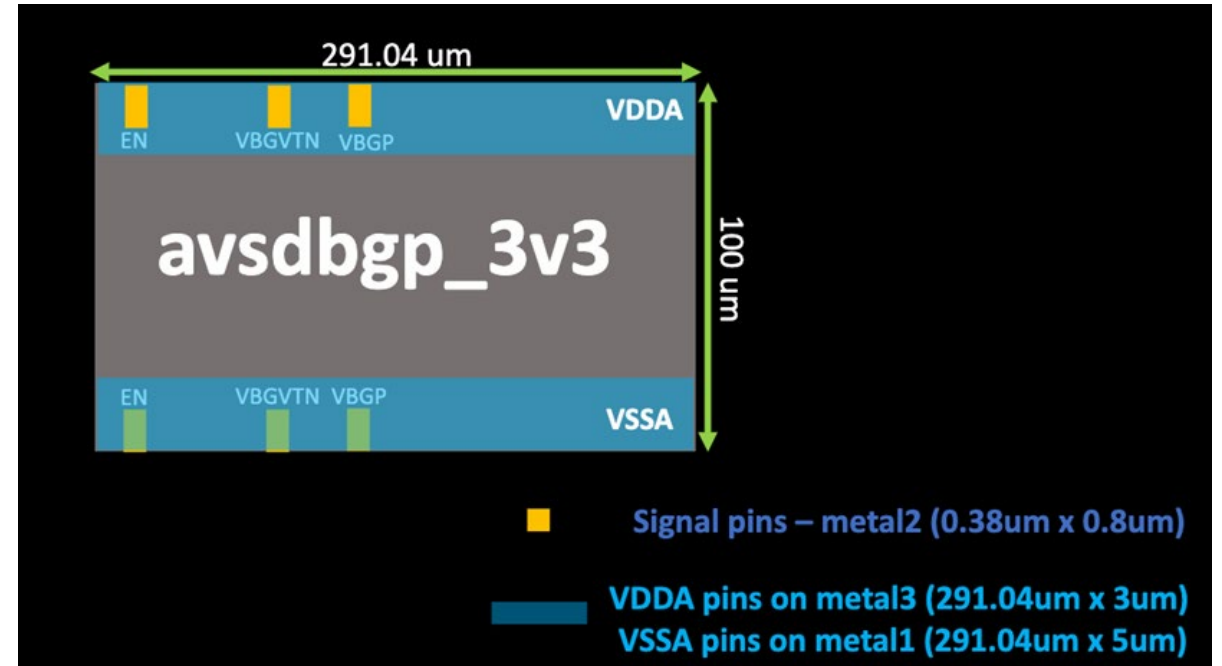
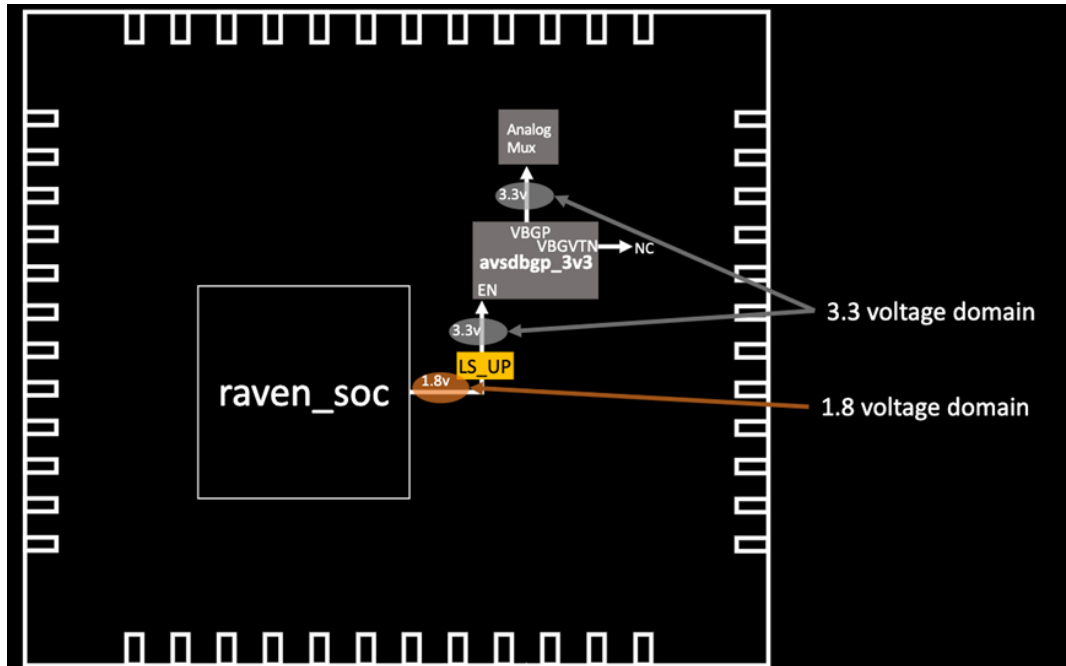


Bandgap Reference Circuit Using Skywater PDK

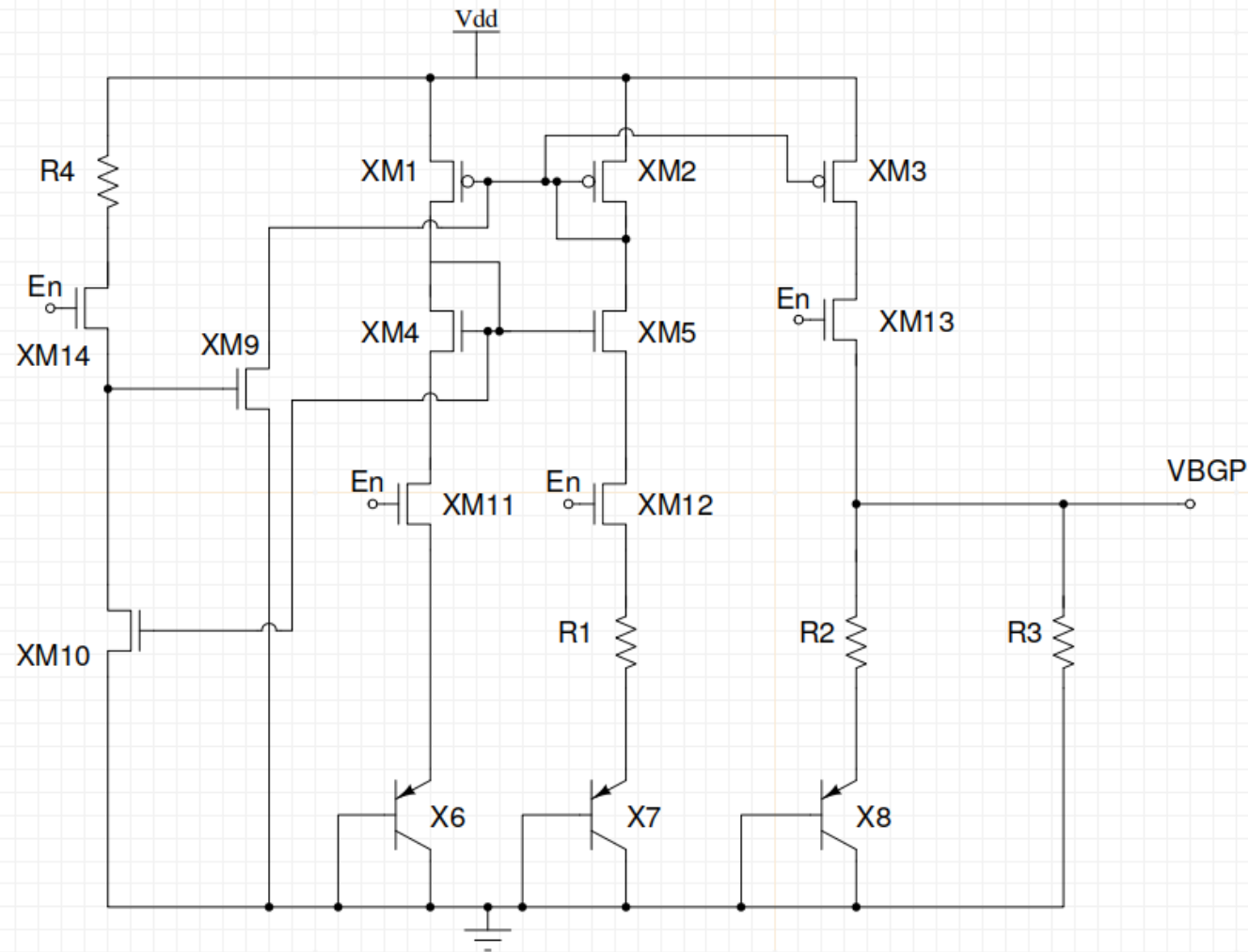
Md Fardin Ahmed

Block Diagram and Specification



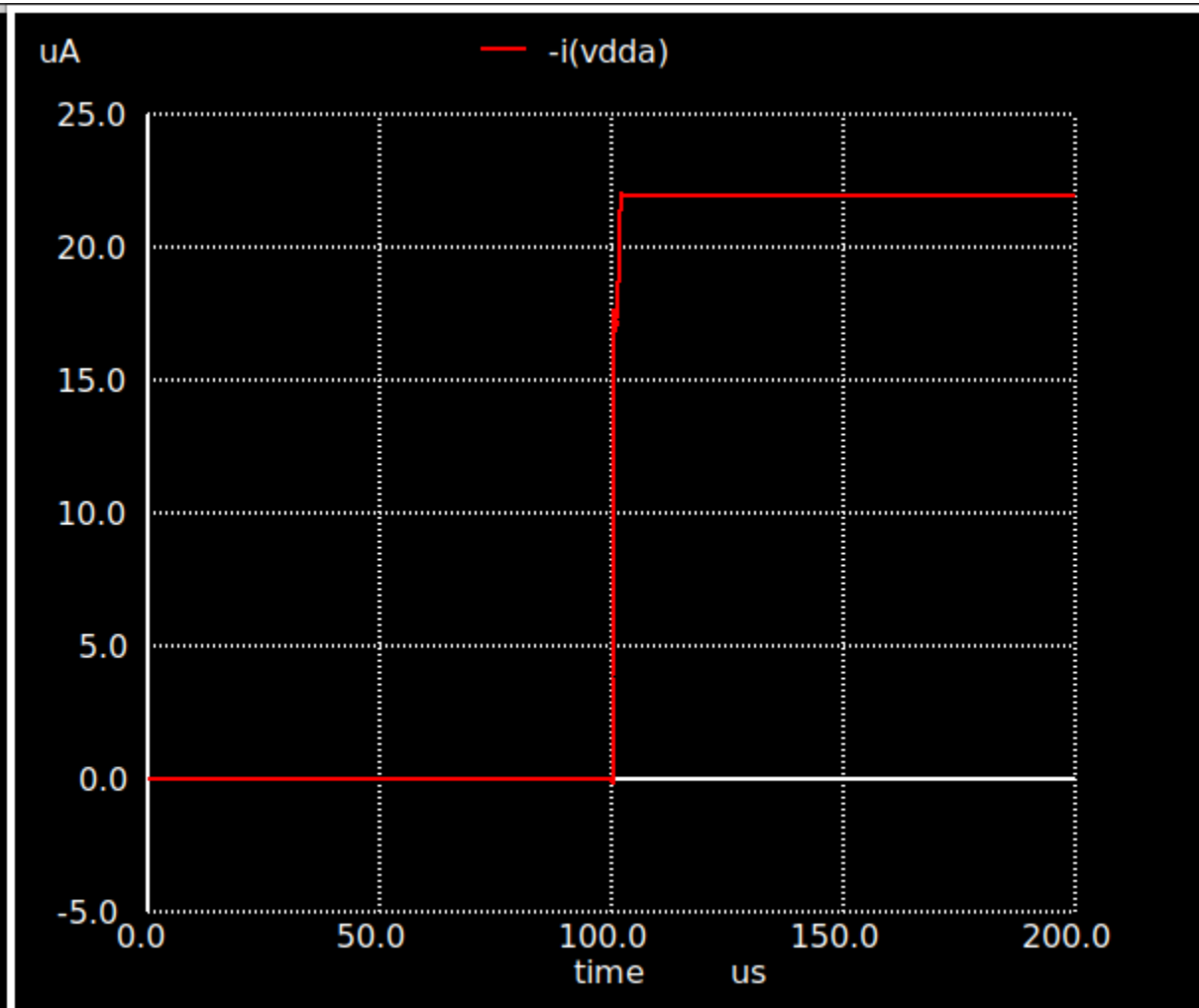
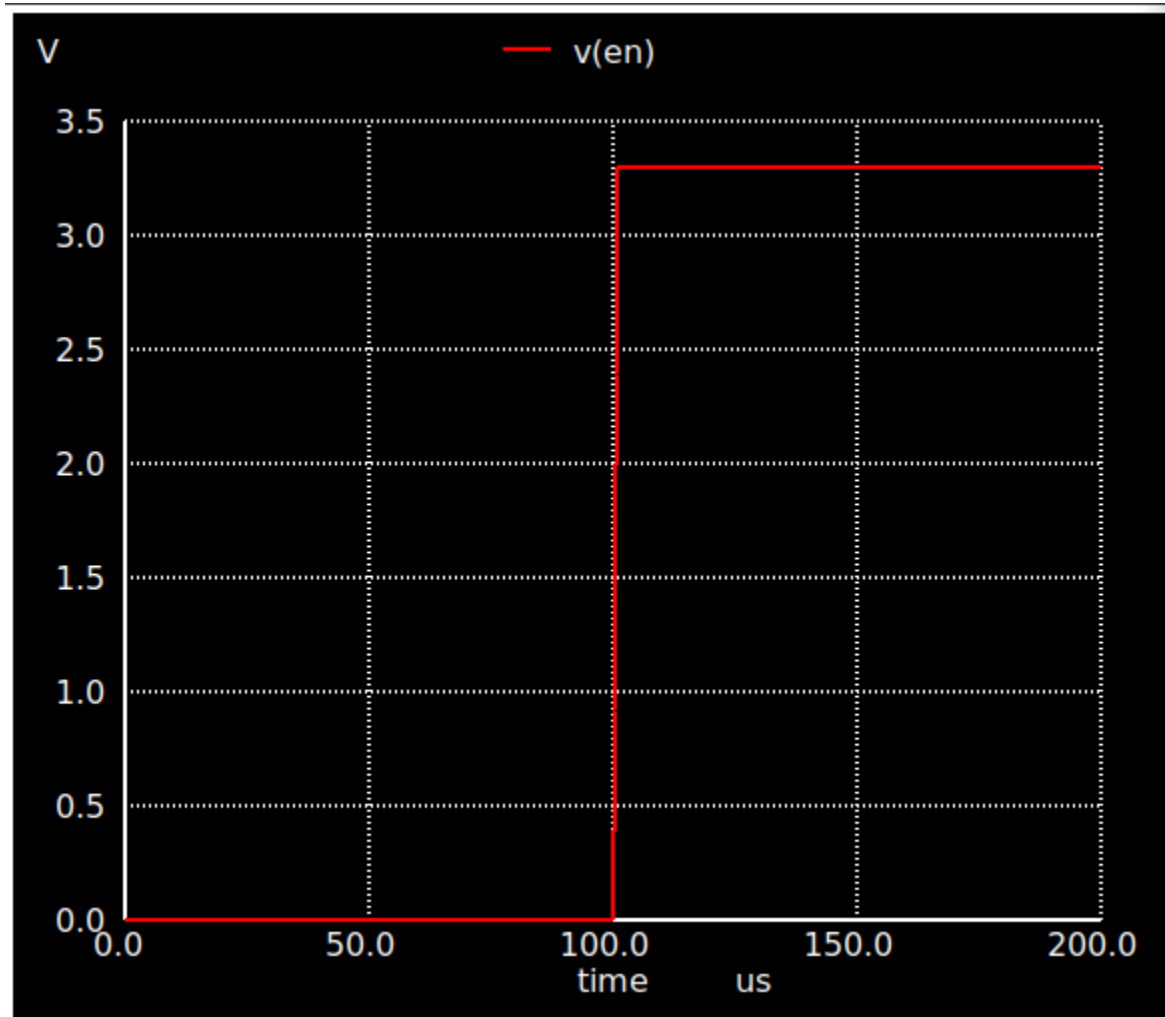
- $V_{ref} = \alpha_1 PTAT + \alpha_2 CTAT$
- $\alpha_1 = R_2/R_1 \ln(n)$
- $\alpha_2 = 1$
- $\alpha_1 \partial(PTAT)/\partial T + \partial(CTAT)/\partial T = 0$

Schematic

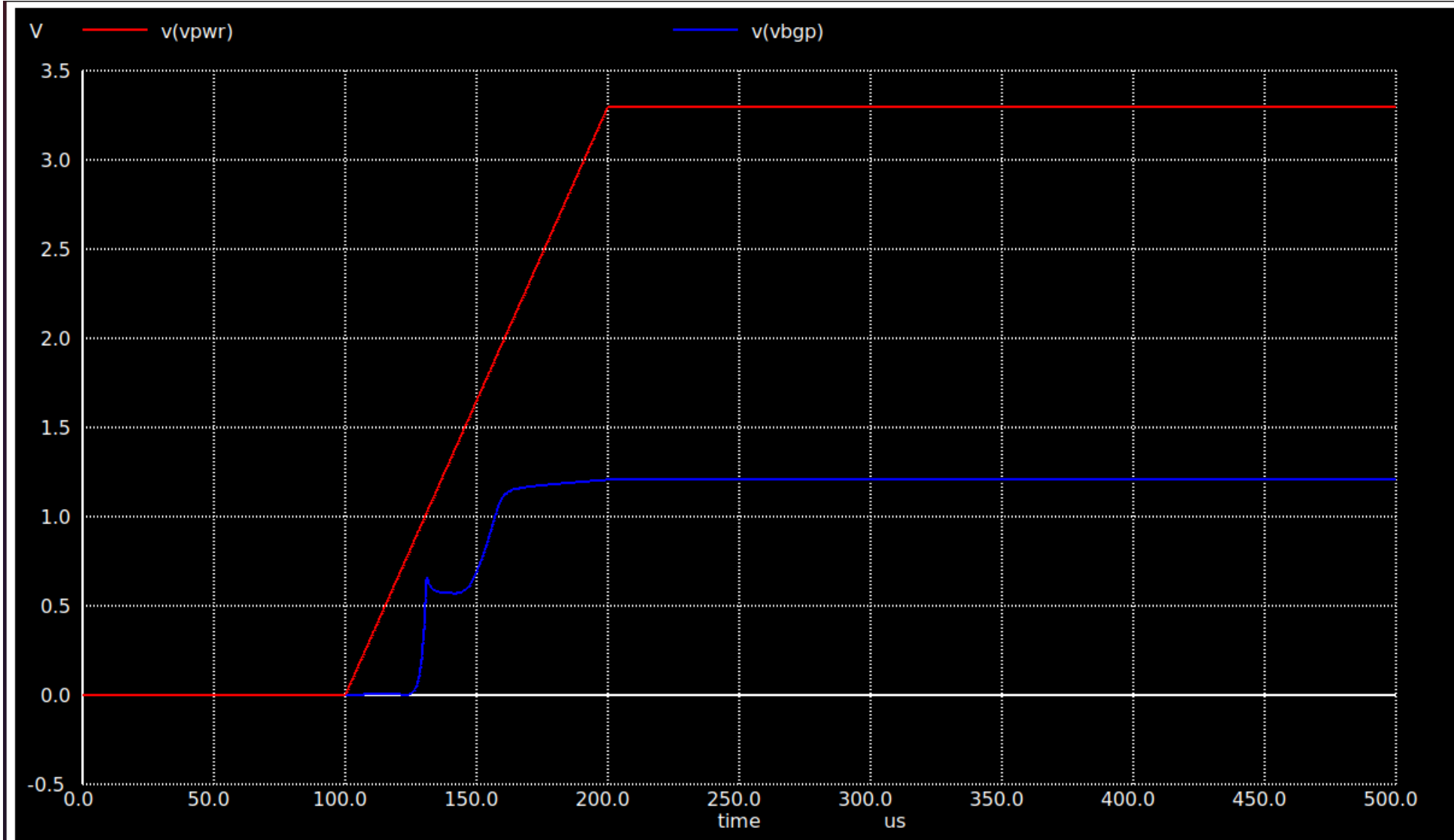


Pre Layout Simulations

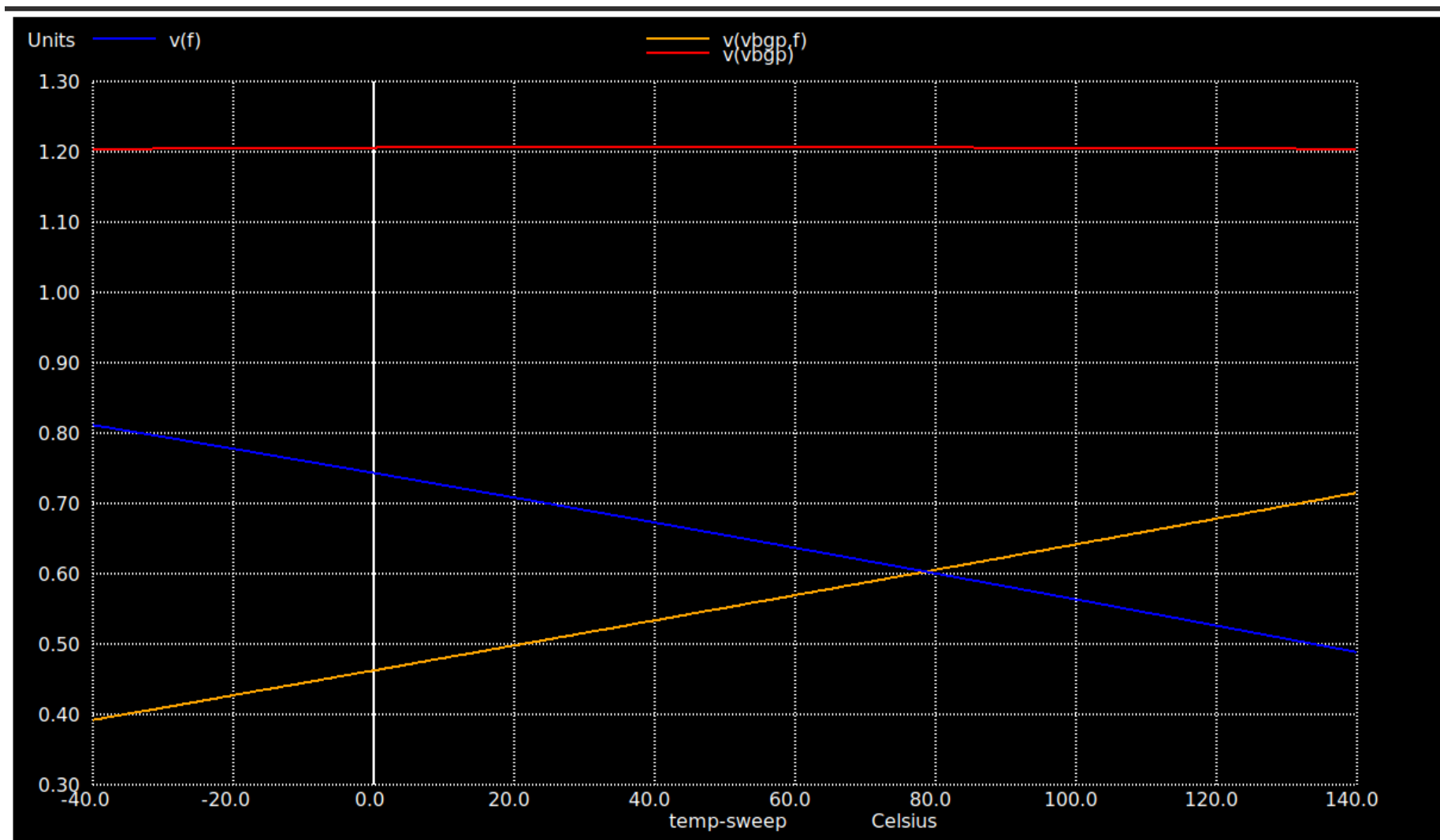
Enable Current



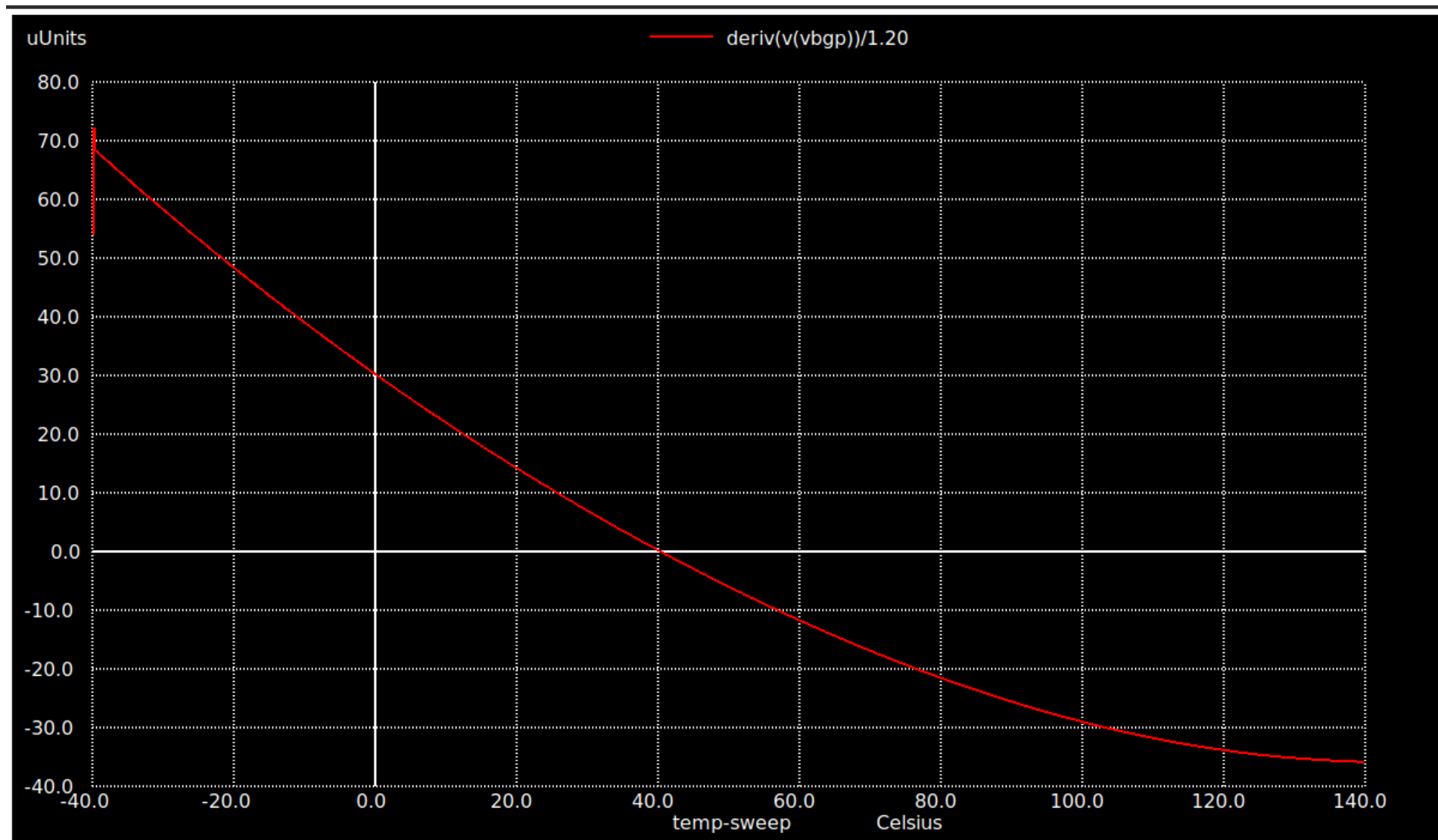
Startup



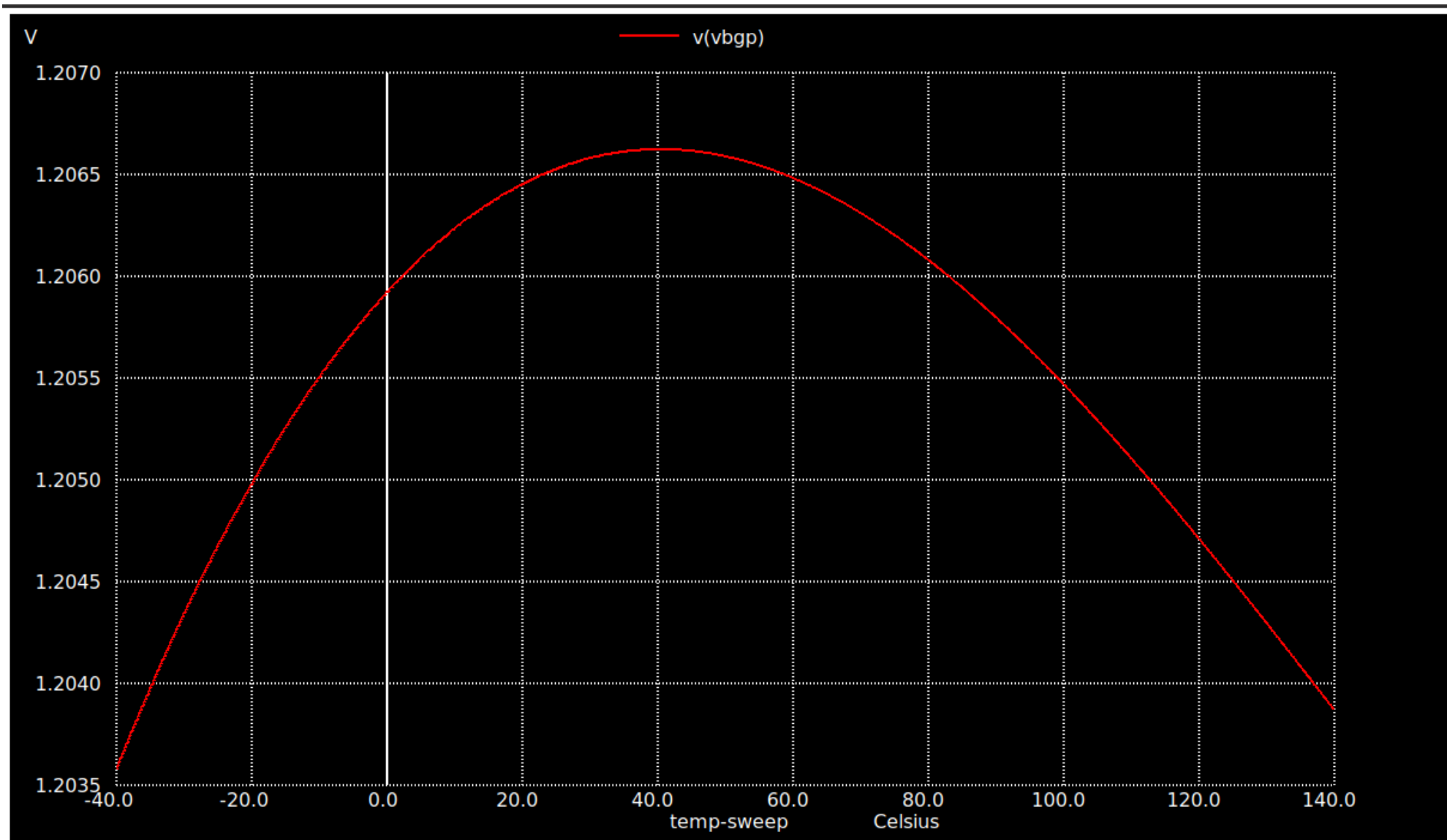
Temp Plot



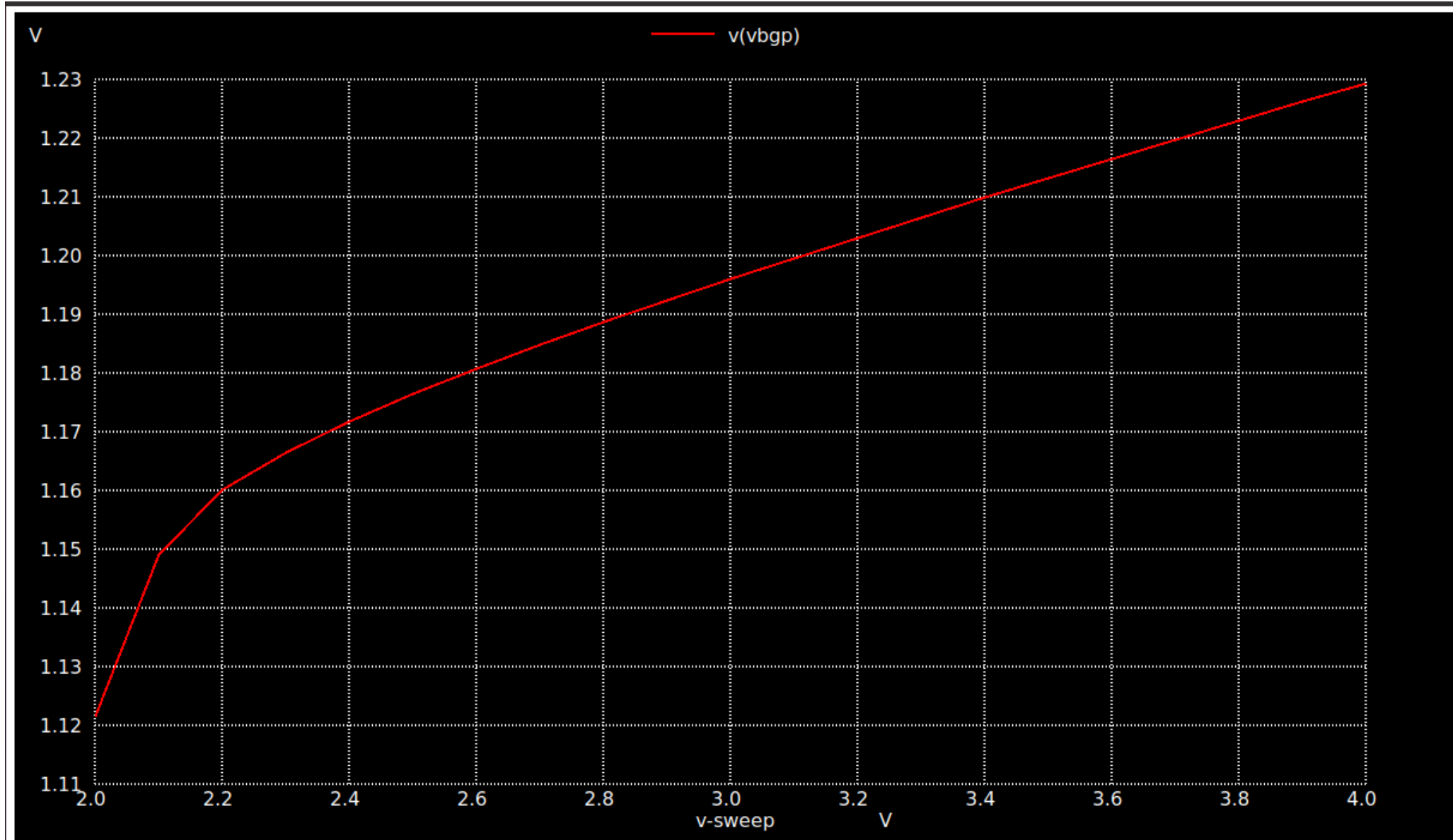
Temp Coefficient



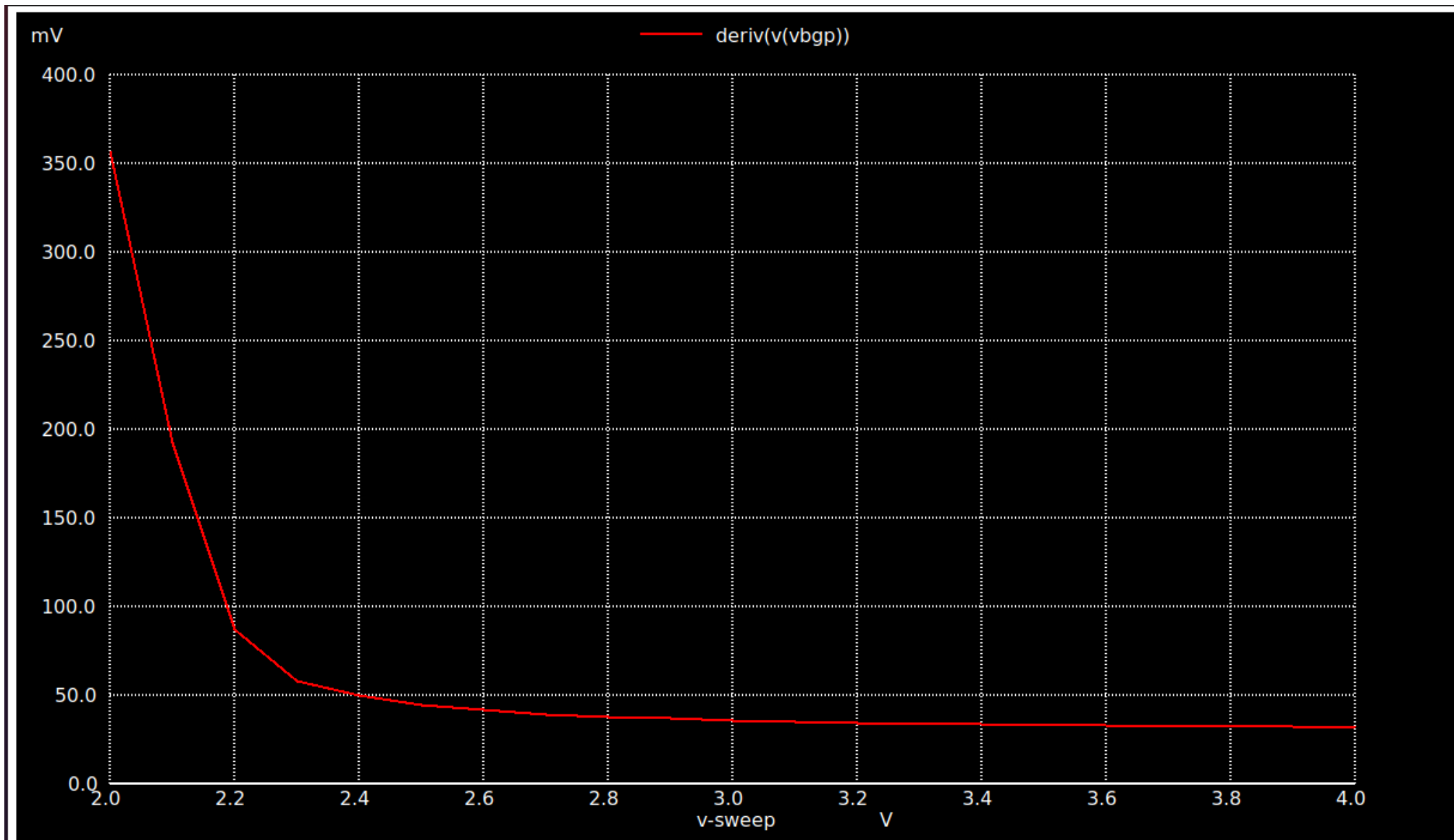
Temp Variation



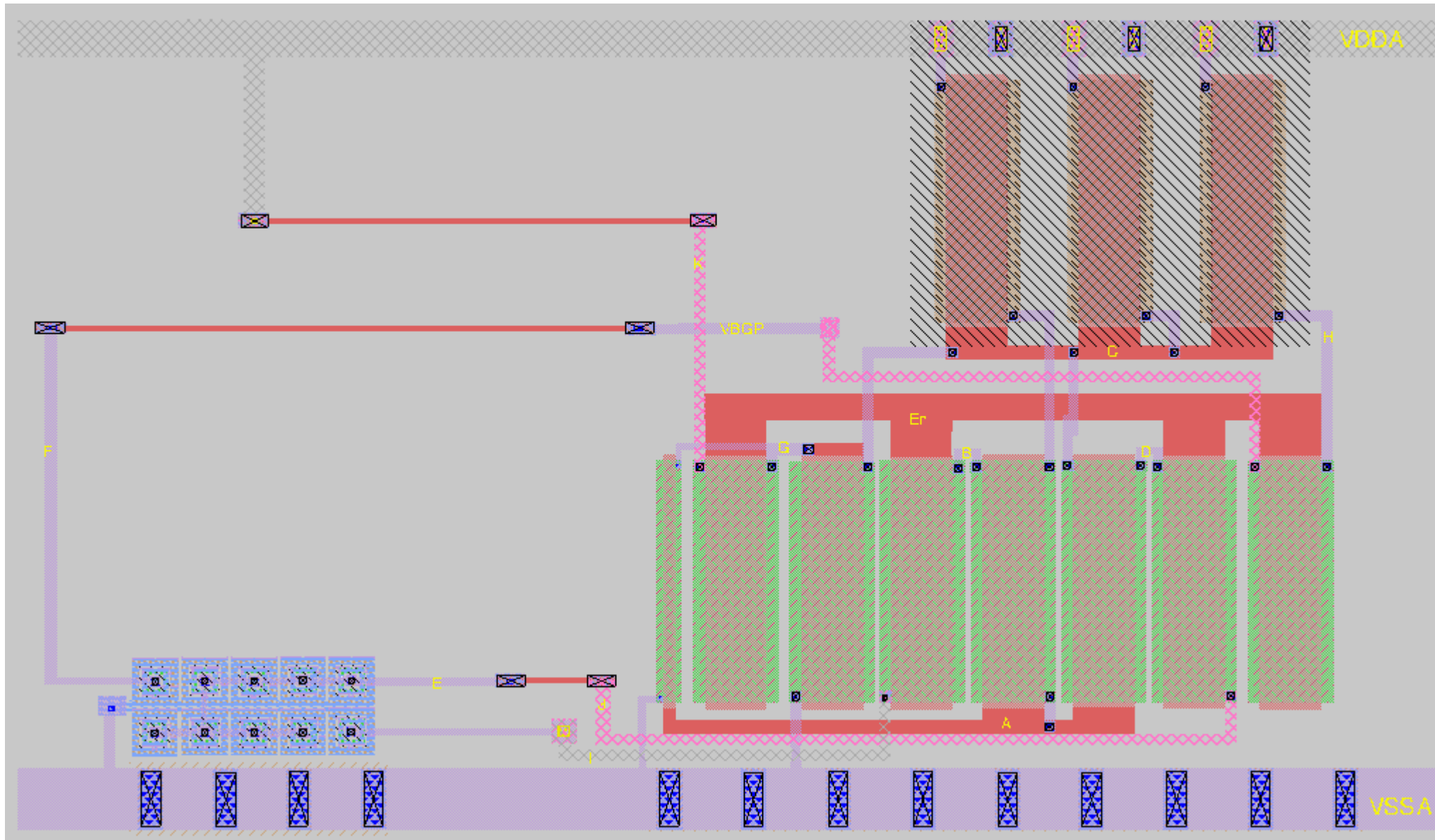
VDD Variation



Vol Variation

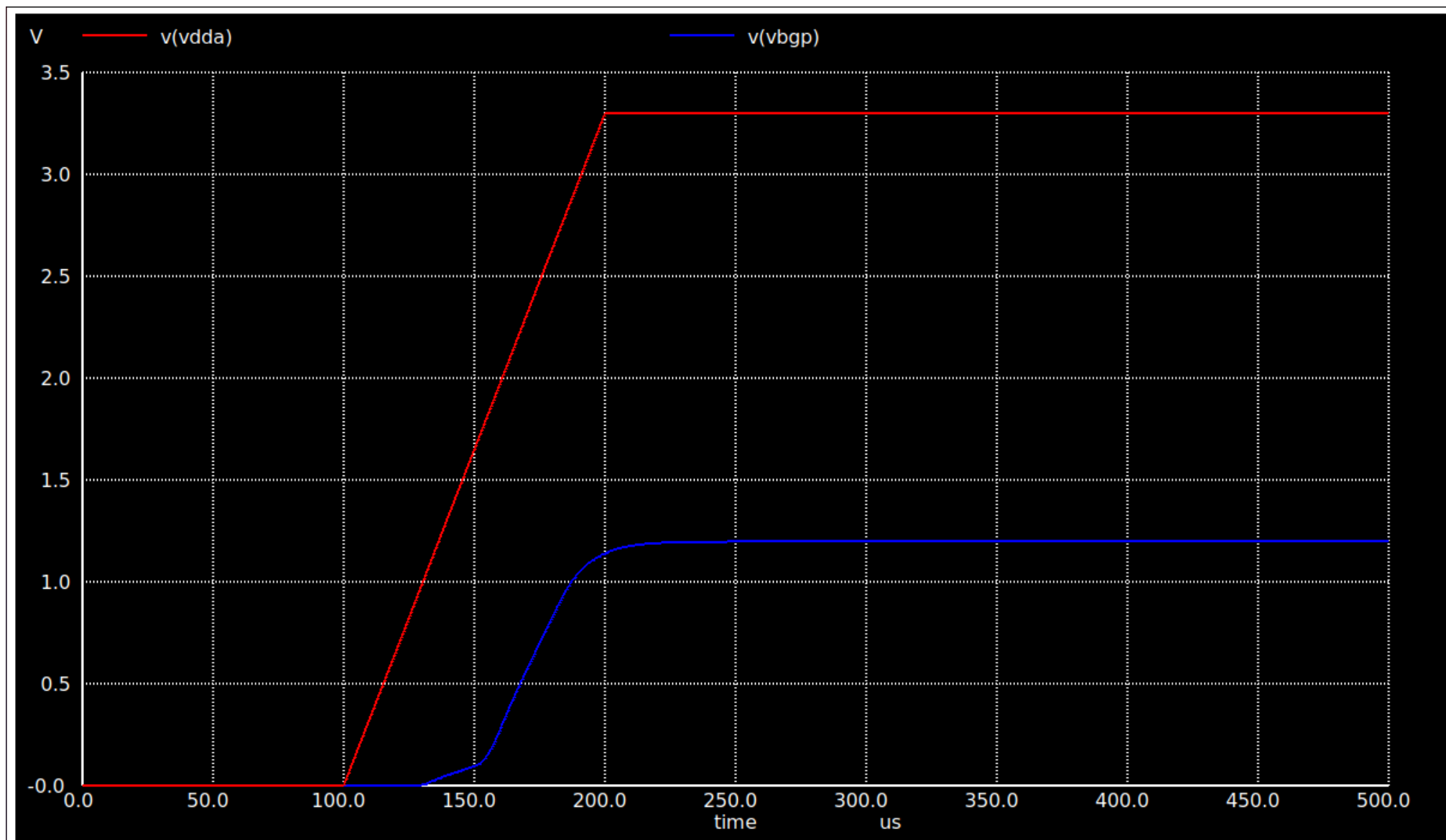


Layout

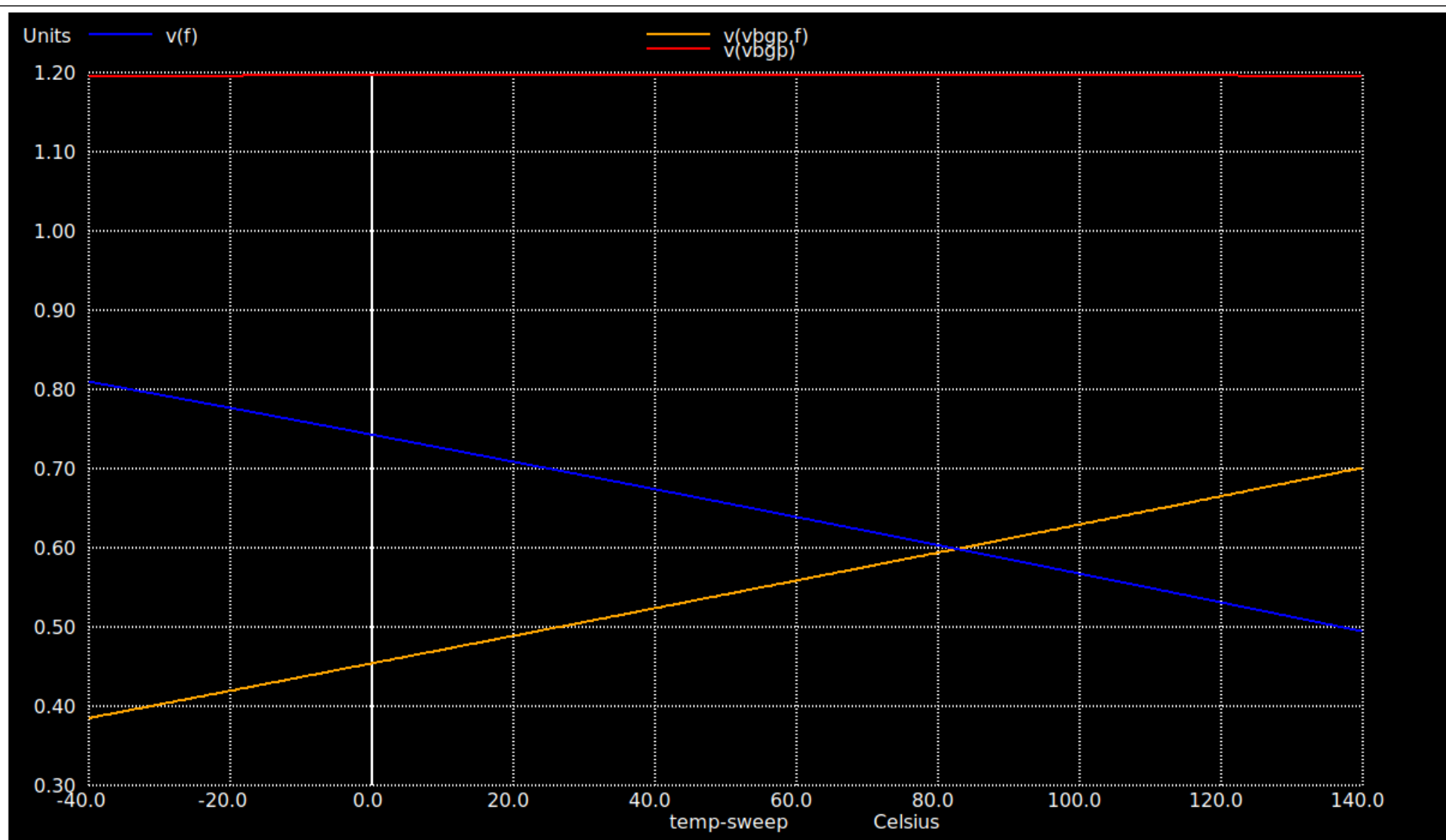


Post Layout Simulations

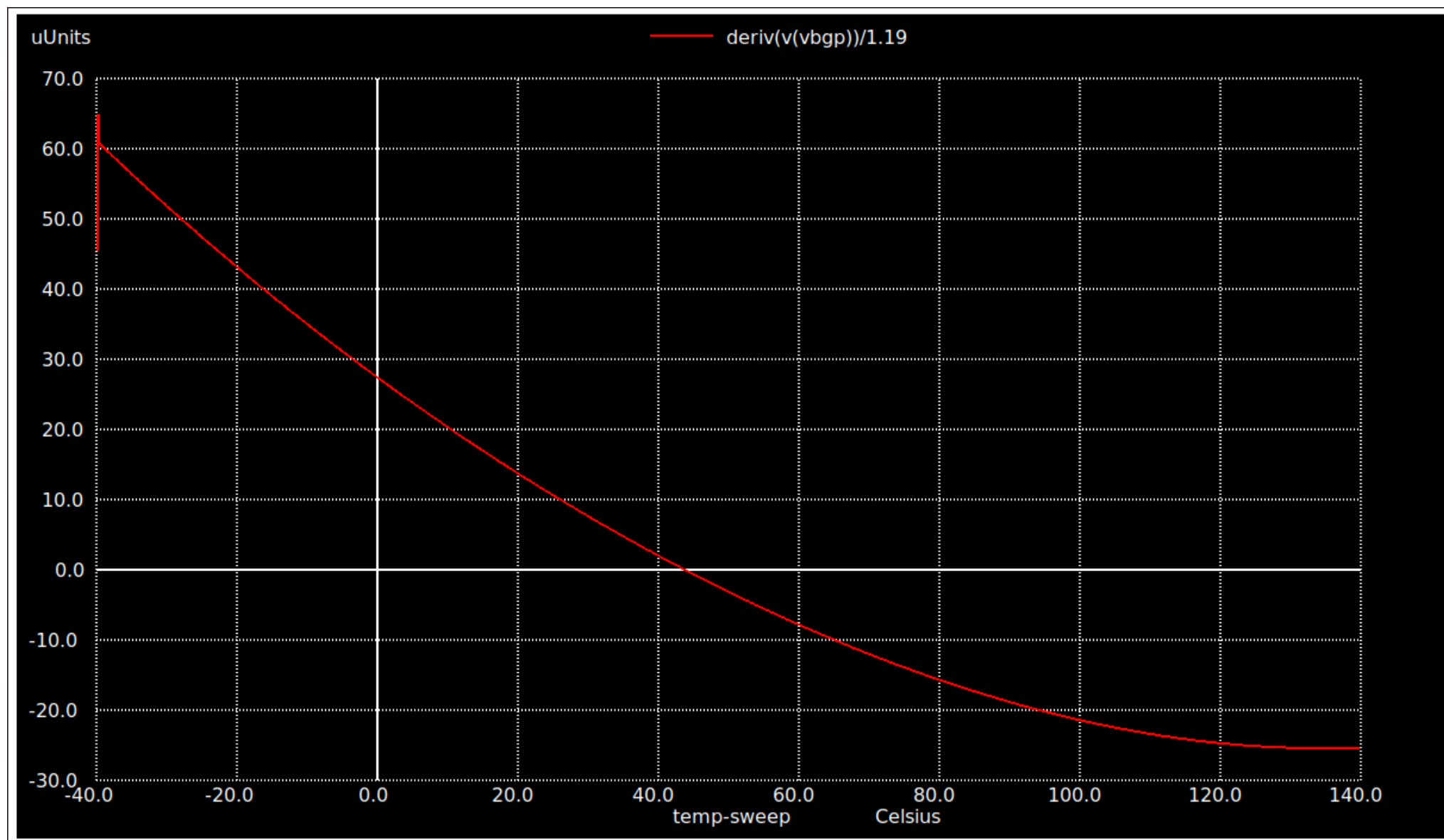
Startup



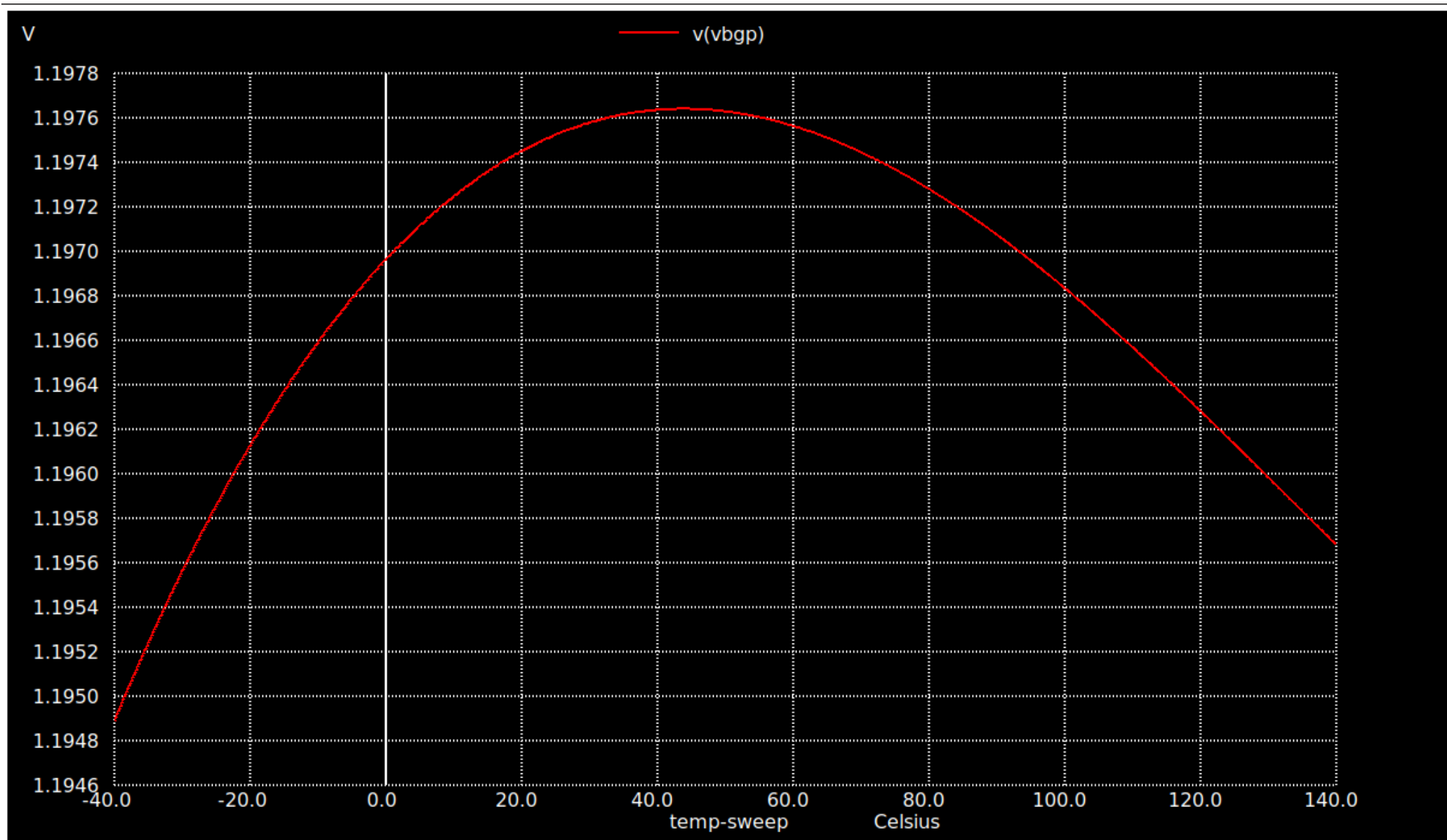
Temp Plot



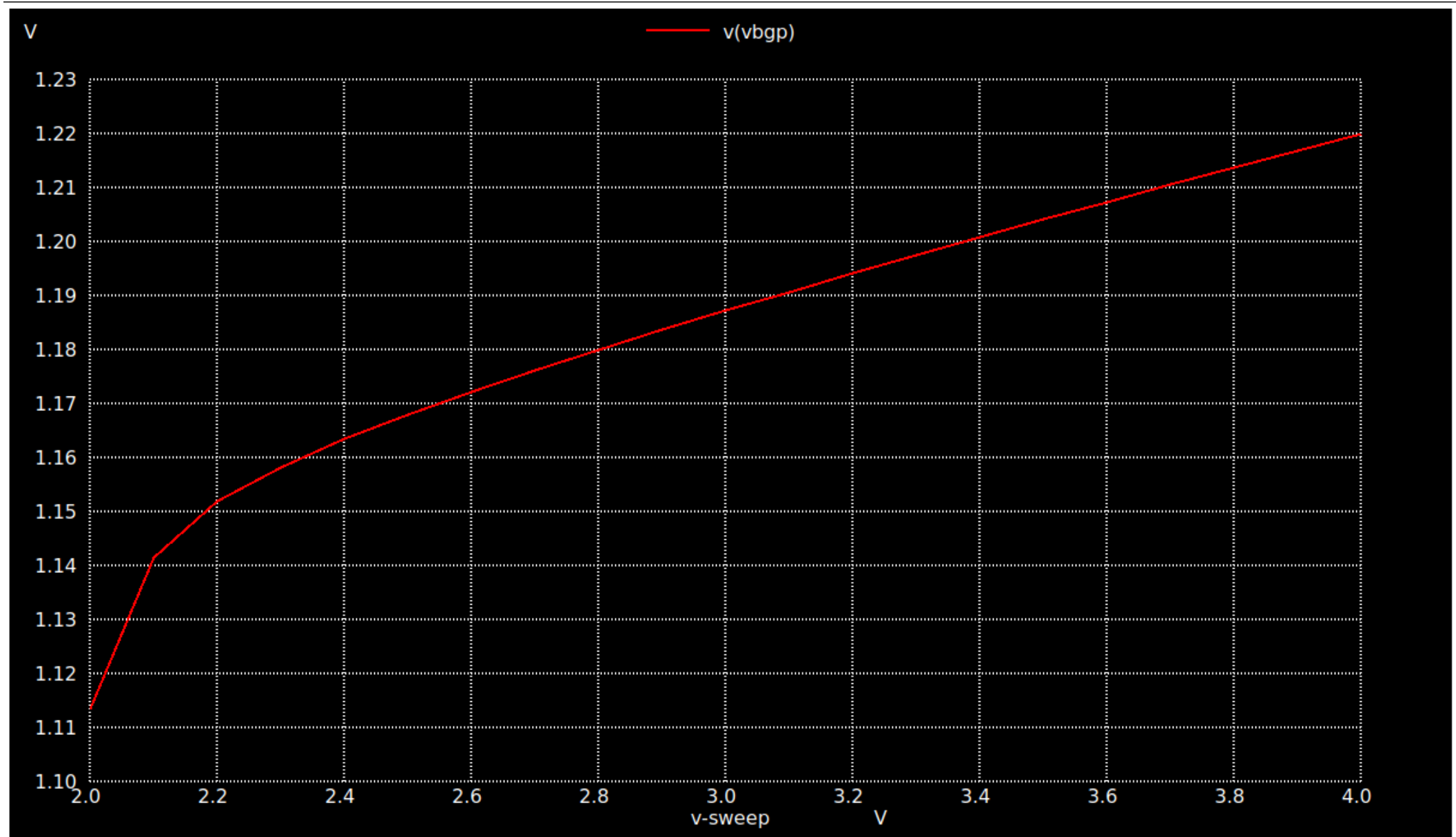
Temp Coefficient



Temp Variation



VDD Variation



Vol Variation

